

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Mastering Semiconductor Key Modules for Enhanced CMOS Process Integration and Device Performance

TGS Ref No: TGS-2024051248

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 9.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	20 August 2026	Complete v7 coverage baseline, eight activities and aligned Lesson Plan.	Dr Alfred Ang
9.0	20 August 2026	Re-aligned to the 139-slide Mobile Photography design-system rebuild, four learning blocks and regenerated activity slide map.	Dr Alfred Ang

TABLE OF CONTENTS

Right-click and update field in Word if page numbers are not shown.

Course Overview

This two-day WSQ course develops the ability to identify CMOS technology process flows, analyse cross-module interactions, determine device and reliability performance, verify integration against product specifications and define accountable follow-up action. The revised deck organises the two official topics into four visual learning blocks.

Field	Details
TSC	Process Integration · ELE-SIS-4002-1.1
Course code	TGS-2024051248
Duration	16 hours over 2 days
Assessment	Day 2, 4:00 PM-6:00 PM (WA 4:00-5:00 PM; PP 5:00-6:00 PM).
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Identify process flows and modules for specific CMOS technologies in manufacturing.
- LO2: Determine CMOS device performance and verify process integration in accordance with product specifications.

Daily Schedule

Day 1 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:00-9:20	20 min	Digital attendance, introductions, outcomes and assessment briefing	Facilitated orientation	Slides 1-18
9:20-10:30	70 min	Block 1A: patterning, etch and fab control	Visual explanation + evidence discussion	Slides 19-38
10:30-10:40	10 min	Morning break	Break	-
10:40-11:25	45 min	Activity 1: CMOS process module map	Guided integration mapping	Slides 39-41
11:25-12:30	65 min	Activity 2: pattern-transfer process window	Workbook analysis + decision review	Slides 42-45
12:30-1:15	45 min	Lunch	Break	-
1:15-2:35	80 min	Block 1B: implant, thermal, films and integrated flow	Visual explanation + cross-section review	Slides 47-66
2:35-2:45	10 min	Afternoon break	Break	-
2:45-3:45	60 min	Activity 3: implant and thermal budget	Evidence analysis	Slides 67-69
3:45-4:45	60 min	Activity 4: film deposition and CMP	Integration review	Slides 70-72
4:45-6:00	75 min	Integrated CMOS flow and Day 1 evidence review	Facilitated synthesis	Slides 73-74

Day 2 Schedule

Time	Duration	Topic / Activity	Method	Slides
9:00-9:15	15 min	Digital attendance and Day 1 retrieval	Recap	-
9:15-10:30	75 min	Block 2A: integration performance, yield and reliability	Visual instruction + evidence review	Slides 75-94
10:30-10:40	10 min	Morning break	Break	-
10:40-11:30	50 min	Activity 5: STI integration review	Flow analysis + activity	Slides 95-97
11:30-12:30	60 min	Activity 6: yield and process capability	Worked example + workbook	Slides 98-101
12:30-1:15	45 min	Lunch	Break	-
1:15-2:15	60 min	Block 2B: SPC, packaging and future integration	Visual instruction + decision models	Slides 103-122
2:15-2:55	40 min	Activity 7: SPC CD excursion response	Evidence analysis + workbook	Slides 123-125
2:55-3:35	40 min	Activity 8: future-technology integration gate	Technology gate review	Slides 126-129
3:35-4:00	25 min	Course synthesis and assessment preparation	Q&A and readiness check	Slides 130-135
4:00-5:00	60 min	Written Assessment (WA-SAQ)	Individual · open book	-
5:00-6:00	60 min	Practical Performance (PP), submission and sign-off	Individual · open book	-

Topic-by-Topic Breakdown

Topic 1 - CMOS Process Modules and Process Flow · Slides 19-73

Learning Block 1A - Patterning, Etch and Fab Control · Slides 19-45

- The Wafer Fab as a System
- Contamination Has Multiple Pathways
- The Lithography Stack
- Mask Tone and Resist Tone
- Resist Coating and Bake
- Resolution, NA and Process Margin
- Overlay, OPC and Mask Protection
- Wet Etch versus Dry Etch
- Etch Selectivity and Profile
- Strip and Clean
- Pattern-Transfer Window
- Patterning Evidence Gate

Activity 1 - Build a CMOS Process Module Map · Slide 39

Organise the legacy module coverage into a defensible FEOL-to-BEOL flow and expose the interfaces that can propagate variation.

Activity 2 - Diagnose a Pattern-Transfer Process Window • Slide 42

Connect mask/resist/exposure decisions to etch profile and select a stable corrective action without masking the root cause.

Learning Block 1B - Implant, Thermal, Films and Integrated Flow • Slides 47-73

- Implant Dose and Energy
- Beamline Risks
- Crystal Damage and Activation
- Diffusion, Furnace and RTP
- Thermal Oxidation
- CVD, PECVD and Epitaxy
- PVD, Barrier and Seed
- Copper Dual Damascene
- CMP Mechanics and Chemistry
- Planarity Failure Modes
- Integrated FEOL and BEOL
- Cross-Module Budgeting

Activity 3 - Balance Implantation and Thermal Budget • Slide 67

Create an implant-to-anneal decision trail that balances activation, junction depth, damage recovery and channeling risk.

Activity 4 - Integrate Film Deposition and CMP • Slide 70

Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.

Topic 2 - CMOS Process Integration and Performance • Slides 75-129

Learning Block 2A - Integration Performance, Yield and Reliability • Slides 75-101

- Isolation and Well Integration
- Transistor Formation
- Yield Is Multiplicative
- Defect Density and Die Area
- Metrology by Question
- Measurement Strategy
- Electrical Performance
- Reliability Evidence
- HCI, TDDB and Electromigration
- Control Limits versus Specifications
- Capability: Cp and Cpk
- Evidence-Based Release

Activity 5 - Review an STI Integration Flow • Slide 95

Produce the process-flow and challenge analysis used as the foundation for Practical Performance Task 1.

Activity 6 - Analyse Yield and Process Capability - Slide 98

Recreate and improve the legacy ProcessCapability exercise using an auditable Excel workbook with dynamic formulas.

Learning Block 2B - SPC, Packaging and Future Integration - Slides 103-129

- SPC as a Feedback System
- Select the Chart from the Data
- Variable Control Charts
- Attribute Control Charts
- Signals Beyond One Point
- Excursion Response
- Packaging Process Flow
- Wire Bond and Flip Chip
- 2.5D, 3D and Chiplets
- FinFET
- GAAFET Nanosheets
- Beyond Today's CMOS

Activity 7 - Respond to an SPC CD Excursion - Slide 123

Recreate the legacy P-chart/SPC exercises and generate the corrective-action chain used in Practical Performance Task 2.

Activity 8 - Run a Future-Technology Integration Gate - Slide 126

Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.

Resources Required

- v9.0 PowerPoint/PDF learner slides
- v9.0 Learner Guide
- Eight self-contained activity folders with printable Markdown/PDF resources
- Pattern-transfer, capability and SPC Excel workbooks
- Projector, whiteboard/A3 paper, calculator and spreadsheet software
- LMS access for courseware and assessment submission

Assessment

Written Assessment (WA-SAQ) - 2 open-ended questions, K1-K2, 1 hour, open book. Practical Performance (PP) - 2 open-ended tasks, A1-A5, 1 hour, open book.

Day 2, 4:00 PM-6:00 PM (WA 4:00-5:00 PM; PP 5:00-6:00 PM).

Minimum 75% attendance is required. Assessment is individual and open book. Learners submit on the LMS and sign the Assessment Summary Record.

Assessment portal: <https://lms-tms.tertiaryinfotech.com/>